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Assignment 4

Section-A

1. Differentiate between Version and Variant ?

→ A version typically refers to a distinct release or iteration of a product, software or system. Versions are usually identified by a unique number or name, indicating a specific stage of development or a set of changes. Successive versions often represent improvements, bug fixes or new features.

A Variant refers to a customized or modified version of a software product. Variants are created to meet specific requirements, cater to different environments, or address particular use cases without developing entirely seperate version of the software.

2. Define effort estimation.

→ Effort estimation in the context of software development refers to the process of predicting the amount of time, resources and personnel that will be required to complete a software project. It is a critical aspect of project planning and management.

3. Discuss Risk management process.

→ Risk management in software engineering is a systematic process that involves identifying, assessing, prioritizing and mitigating risks that could impact the success of a software project. The goal of risk management is to proactively address potential issues before they can negatively affect the projects schedule, budget or quality.

4. What do you mean by project planning ?

→ Project planning is a crucial part of project management

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focused on creating a detailed plan that outlines the steps and resources necessary to achieve the project's objectives, including identifying the project's scope, establishing a timeline, assigning tasks and resources and budgeting for the project.

5. Define Software size estimation.

→ Software size estimation is an activity in software engineering that is used to determine or estimate the size of a software application or component in order to able to implement other software project management activities.

Section - B

1. Explain need of software project management also role of software project manager in software project.

→ Software Project Management (SPM) is a proper way of planning and leading software projects. It is a part of project management in which software projects are planned, implemented, monitored and controlled.

It is necessary for an organization to deliver quality products, keep the cost within the client's budget constraints and deliver the project as per schedule. Hence, in order software project management is necessary to incorperate user requirements along with budget and time constraints.

A project manager is a character who has the overall responsibility for the planning, design, execution, monitoring, controlling and closure of a project.

Role of a project manager:

(i) Leader: A project manager must lead his team and

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should provide them direction to make them understand what is expected from all of them.

(ii) Medium : The project manager is a medium between his clients and his team. He must coordinate and tranfer all the appropriate information from the clients to his team and report to the senior management.

(iii) Mentor: He should be there to guide ~~the~~ his team at each step and make sure that the team has an attachment. He provides a recommendation to his team and points them in the right direction.

2. Discuss project scheduling in details.

→ Project scheduling simply means a mechanism that is used to communicate and know about that tasks are needed and has to be done or performed and which organizational resources will be given or allocated to these tasks and in what time duration or time framework is needed to be performed. Scheduling in project management means to list out activities, deliverables and milestones within a project that are delivered. To schedule the project plan, a software project manager wants to do the following:

(i) Identify all the functions required to complete the project.

(ii) Break down large functions into small activities.

(iii) Determine the dependency among various activities.

(iv) Establish the most likely size for the time duration required to complete the activities.

(v) Allocate resources to activities.

(vi) Plan the beginning and ending dates for different activities.

(vii) Determine the critical path. A critical way is the group of activities that decide the duration of the project.

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3. Why the project delayed explains in your words.

→ Project delays can occur for a variety of reasons, and they often result from a combination of factors. Here are some explanations for project delays:

(i) Scope Changes: Changes in project scope, requirements, or objectives can lead to delays as the team needs to adjust plans and accomodate the new elements. Unclear or evolving requirements can contribute to scope changes.

(ii) Resource Constraints: Insufficient resources can slow down progress. Without the necessary resources, tasks take longer to complete.

(iii) Technical challenges: Unforseen technical issues, difficulties in implementation, or unexpected obstacles can arise during development, requiring additional time to overcome and resolve.

(iv) Poor Planning: Inadequate project planning can lead to delays. A lack of proper planning can lead to oversights and unanticipated issues.

(v) Dependencies: Projects often have ~~dependicies~~ dependencies on other tasks, teams or external factors. Delays in dependent tasks can create bottlenecks, causing delays throughout the project.

Section-C

1. Discuss organization and team structure. Explain Maslow's hierarchy of needs.

→ Organization Structure: It refers to the framework that defines how tasks are divided, grouped and coordinated within an organization. The structure determines the reporting relationships, communication channels, and decision - making processes. Common types of organizational structure includes Functional structure, Divisional structure, Matrix structure, Flat structure, Hierarchical structure, Network structure.

Team Structure : It defines how individuals within a team are organized to achieve specific goals. It involves determining roles, responsibilities, and communication channels within the team. Types of team structure include: Functional teams, ~~cr~~ cross- functional teams, self- Directed teams, Virtual teams, Project teams.

Maslow's Hierarchy of Needs : It is regarded as one of the most popular theories on motivation. It is a theory of psychology that explains that humans are highly motivated in order to fulfill their needs, which is based on heirarchical order. It is based on a heirarchy of needs, which starts with the most basic needs and subsequently moves on to higher levels. The main goal of this need hierarchy theory is to attain the highest position or the last of the needs i.e., need for self actualization.

Levels of Hierarchy: The levels of Hierarchy in Maslow's need hierarchy theory appear in the shape of a pyramid, where the most basic need is placed at ~~the~~ bottom while the most advanced level of heirarchy is at the top of the pyramid. The various levels in Maslow's hierarchy of needs :

(i) Physiological needs: The physiological needs are regarded as the most basic of the needs that humans have. These are needs that are very crucial for our survival. Eg. Food, shelter, water etc.

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(ii) Safety needs: Once the basic needs of food, shelter, water etc are fulfilled, there is an innate desire to move to the next level. The next level is known as safety needs. Here the primary concern of the individual is related to safety and security.

(iii) Social needs: It is the third level in the need heirarchy theory. It is that stage where an individual having fulfilled his physiological needs as well as safety needs seeks acceptance from others in the form of love, belongingness. Ex- Friendship, Family etc.

(iv) Esteem needs: This is considered as the fourth level of the heirarchy of needs Theory. It is related to the need of a person being recognised in the society. It deals with getting recognition, self respect in the society.

(v) Self-Actualization needs: This is the final level of the theory of heirarchy of needs as proposed by Maslow. It is the highest level of needs and is known as the self-actualization ~~as~~ needs. It relates to the need of an individual to attain or realise the full potential of their ability or potential.

2. What are various goals and objective of SCM? why it is requited. and what are benefits of using it.

→ • System Configuration Management (SCM) is an arrangement of exercises which controls change by recognizing the items for change, setting up connections between those things, making / characterizing instruments for overseeing diverse variants, controlling the changes being executed in the current framework, inspecting and revealing / reporting on the changes made.

The key objectives of SCM are to:

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1. Control the evolution of software systems: SCM helps to ensure that changes to a software system are properly planned, tested and integrated into the final product.
2. Enable collaboration and coordination: SCM helps teams to collaborate and coordinate their work, ensuring that changes are properly integrated and that everyone is working from the same version of the software system.
3. Provide version control: SCM provides version control for software systems, enabling teams to manage and track different versions of the system and to revert to earlier versions if necessary.
4. Faciliate replication and distribution: SCM helps to ensure that software systems can be easily replicated and distributed to other enviroments, such as test, production and customer sites.
5. SCM is a critical component of software systems, and effective SCM practices can help to improve the quality and reliability of software systems as well as increase efficiency and reduce the risk of errors.

Benefits of using SCM are:

(i) Improved productivity and efficiency by reducing the time and effort required to manage software changes.

(ii) Reduced risks of errors and defects by ensuring that all changes are properly tested and validated.

(iii) Increased collaboration and communication among team members by providing a central repository for software artifacts.

(iv) Improved quality and stability of software systems by ensuring that all changes are properly controlled and managed.

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3. Write a short note on the following :-

a) PERT.

b) GANTT CHART

→ a) PERT (Programme Evaluation Review Technique) :

It is procedure through which activities of a project are represented in its appropriate sequence and timing. It is a scheduling technique used to schedule, organize and integrate tasks within a project. PERT is basically a mechanism for management planning and control which provides blueprint for a particular project. All of the primary elements or events of a project have been finally identified by the PERT. In this technique, a ~~pert~~ PERT chart is prepared which represent a schedule for all the specified tasks in the project. The reporting levels of the tasks or events in the PERT Charts is somewhat same as defined in the work breakdown structure (WBS).

b) GANTT Chart (Generalized Activity Normalization Time table) :

It is a type of chart in which series of horizontal lines are present that show the amount of work done or production completed in given period of time in relation to amount planned for those projects. It is horizontal bar chart developed by Henry L. Gantt in 1917 as production control tool. It is simply used for graphical representation of schedule that helps to plan in an efficient way, coordinate and track some particular tasks in project. The purpose of GANTT chart is to emphasize scope of individual tasks. Hence set of tasks is given to Gantt chart as input. Gantt chart also known as timeline chart.